

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			Hess' cycle drawn with all arrows in correct direction / $\Delta H = \Delta H \text{ reactants} - \Delta H \text{ products}$ (1) correct values -1300, -572 and -1600 on arrows / $\Delta H = [-1300 + (-572)] - (-1600)$ (1) $\Delta H = -272 \text{ kJmol}^{-1}$ (1)		3		3	1 1	
	(c)			bond enthalpy terms are averages / same bond in different environment will have slightly different bond enthalpy			1	1		
	(d)			(electrophilic) addition / hydrogenation	1			1		
				Question 9 total	5	6	1	12	3	0